جامعة مصر للمعلوماتية
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Egypt University of Informatics
Computer and Information Systems
Data Analysis Course

The Analysis of the Performance of Data Analysis Students

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Introduction

The “starving student” has been updated to become the “exhausted student” in an age where there is no respite from constant digital connectivity and heavy academic burdens. While we view ourselves as machines that demand peak brain function without considering that which fuels us, namely sleep and exercise.

However, does adding another hour of TikTok browsing prevent you from concentrating? Is going to the gym really the cause of you being more productive when you study? Herein lies the purpose of this paper; rather than conjecture, it presents a concrete analysis of how our habits directly affect our cognitive functions through data.

Research Question

Physical Activity and Its Impact on Energy Concentration

Hypothesis

Population of Interest:

All students (either university or school)

Sampling Method:

The sample method used is convenience sampling (aka the “Honest Approach). This sampling method is the best approach for our research since it ensures many participants and data from all majors and years. Since the variables (cell phone use, sleeping hours, and focus) are very subjective, the online survey helps obtain a more truthful than conventional techniques would. The volume and truthfulness of information obtained are vital in proving the correlation between exercise and brain function.

Bias Identification:

Our bias is the belief that digital habits are the main cause of student exhaustion and that exercise is the best 'fuel' for focus. We went into this project expecting to prove that phones are the problem and activity is the solution. To make sure our opinion didn't influence the results, we kept the questions strictly objective. We asked for actual hours and days rather than asking students if they 'felt' distracted or 'felt' healthy. By using a neutral 1–10 rating scale for concentration, we ensured we werent 'leading' the participants to give us the answers we expected to see.

Survey Questions:

1) **How many days per week do you engage in physical activity?**
2) **On average, how would you rate your daily energy level?**
3) **On average, how would you rate your concentration while studying?**
4) **On average, how many hours do you sleep per night**
5) **How many hours per day do you spend using your phone?**

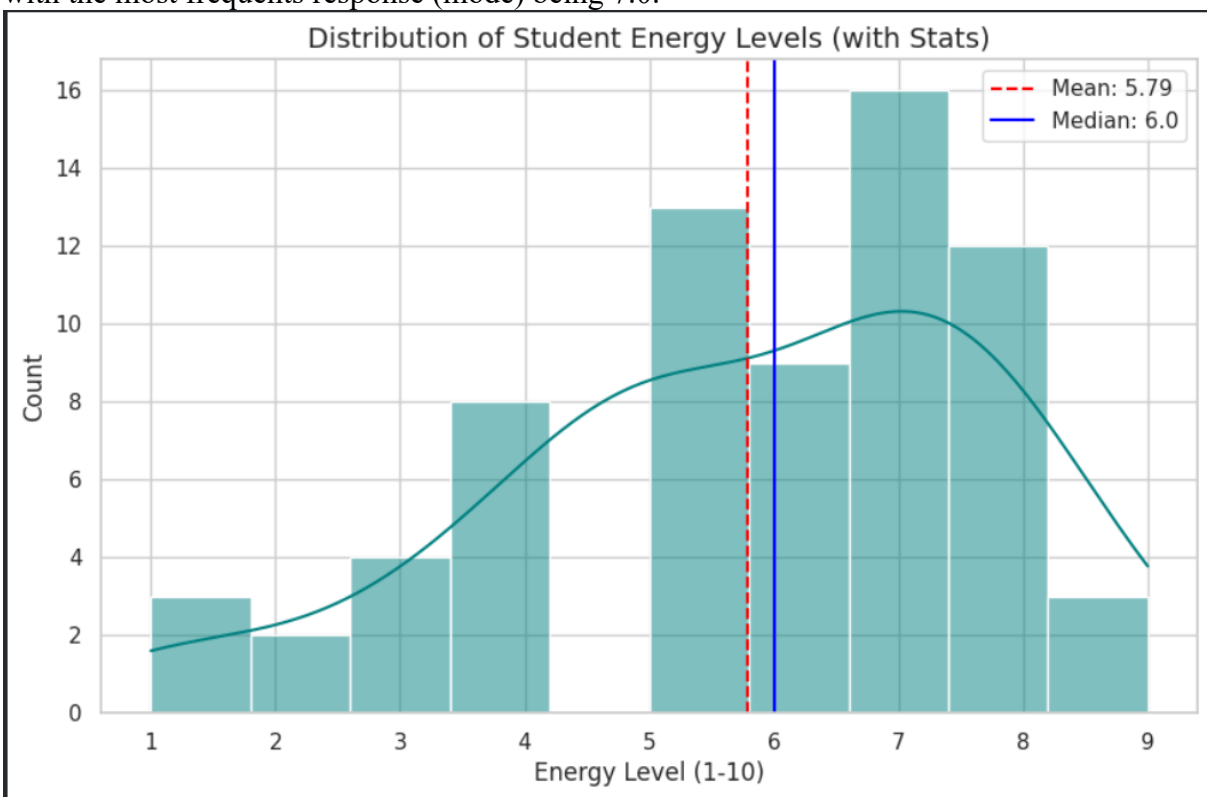
6) **How many hours do you study per day?**

Online survey link:
https://docs.google.com/forms/d/e/1FAIpQLScOVjn0d1GAmIZDgumI4FxUsuLg9D4oZN2uoE1XHoc7ILzVQQ/viewform?usp=publish-editor

Number of samples collected: 70

Analysis:

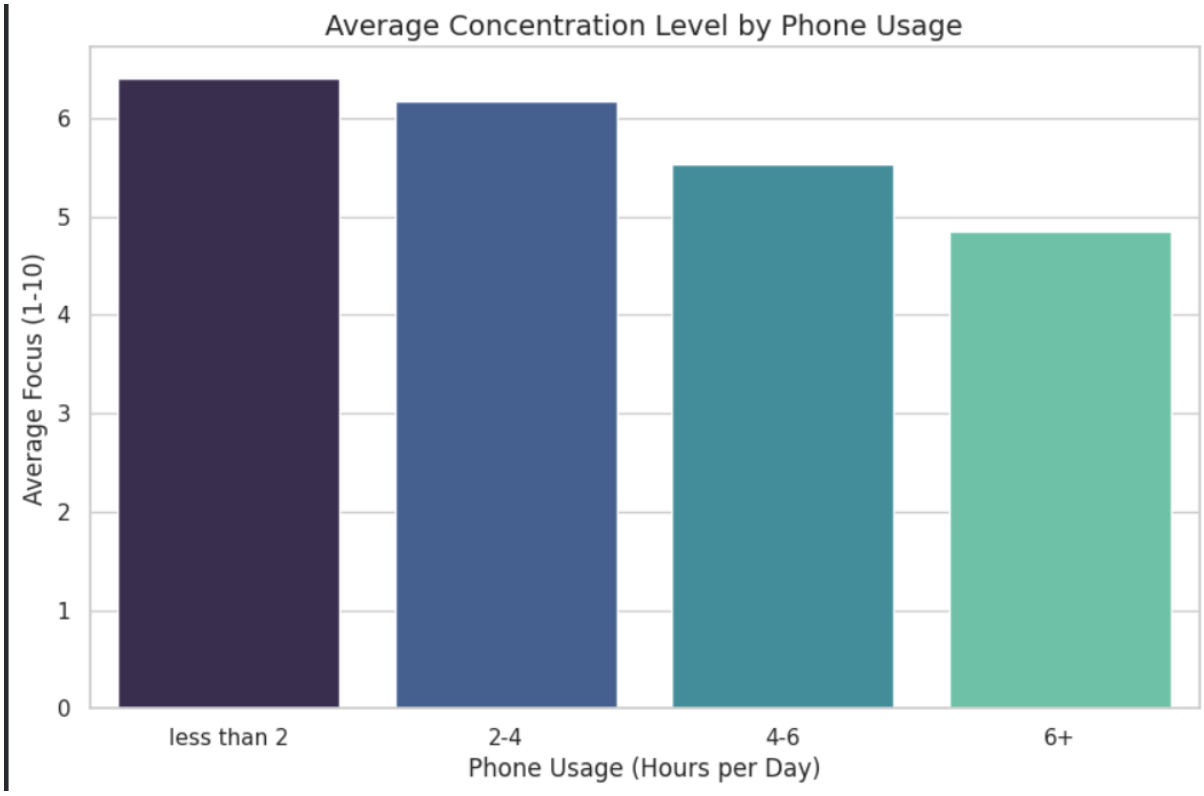
- **Energy levels:** The numerical stats show a mean of 5.79 and a median of 6.0, with the most frequents response (mode) being 7.0.



The bar chart shows how students rated their energy levels on a scale, with a density curve showing the overall trend of the data. Since the median ad mode are higher than the mean, we can conclude that while most students feel moderately energetic, a few very low scores are slightly pulling the average down

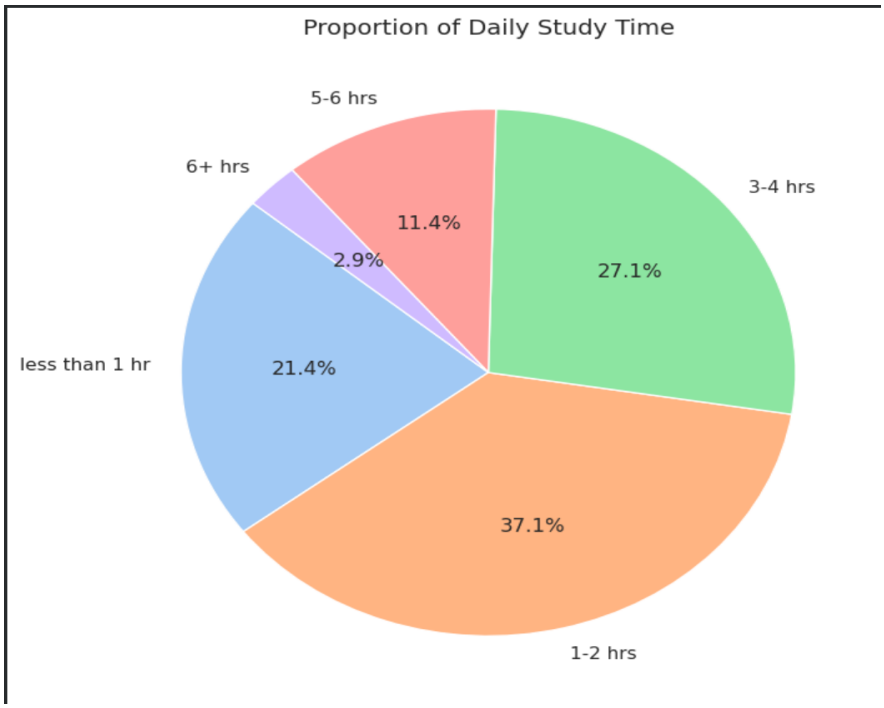
- **Physical activity:** the mode is 1-2 hours per week. The data shows that most participants are physically active but stay within a light-to-moderate range, which represents the standard fitness level

- **Impact of Phone usage:** The statistics show a mean focus of 5.56 and a mode of 6.0. The mode for phone usage is 4-6 hours.



The bar chart compares different levels of phone usage against the average concentration levels reported by students. The analysis shows that students in the highest phone usage categories tend to report focus scores near or below the mean of 5.56. This supports the conclusion that high digital engagement likely acts as a ceiling for student concentration levels.

- **Daily study time:** mode is 1-2 hours



This pie chart breaks down the percentage of students that fall into different study duration categories. We can conclude that the largest segment of students' studies for 1-2 hours daily, which aligns closely with the most common amount of time spent on physical exercise.

Conclusion

Based on statistical relationship between phone usage and focus, we can evaluate the impact of digital habits on student concentration. The data indicates an inverse relationship between screen time and concentration levels. While the overall Mean Focus is 5.56, students in the "4-6 hours" phone usage category (the Mode) tend to report focus scores that align with or fall below this average. Consequently, we can conclude that as phone usage moves toward the higher recorded ranges, we can see that too much phone usage is keeping students from concentrating well, suggesting that high phone dependency may act as a limiting factor for peak mental concentration. The data indicates that while students are functioning adequately, there is a clear opportunity to improve focus and energy levels by redistributing time from high-usage digital activities toward increased physical exercise and sleep. This balanced approach would likely push the current average focus score of 5.56 toward a higher, more optimal range.

Any potential issues

It is important to remember that this data came from self-reporting. This means students might have guessed their times or given themselves higher scores to look better. Also, because we used ranges (like "1-2 hours"), we don't know the exact minutes spent, which makes the data a little less precise. However, the overall "big picture" clearly shows that high phone use and moderate exercise are the two biggest factors shaping student life today.